

Beyond the Needle: Stress-Testing Long-Context Intelligence

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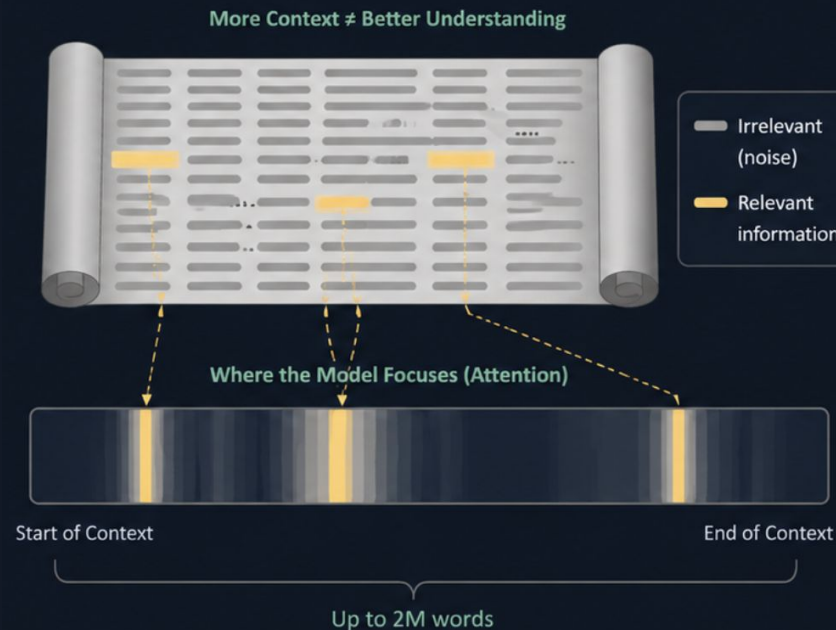
The Long-Context Comprehension Gap

Core Question

Do larger context windows guarantee better comprehension? We evaluate whether structured prompting improves or degrades performance on LongBench v2 — a 503-instance benchmark spanning six task categories with contexts up to 2M words.

Why It's Hard

- Context capacity $\neq$ context utilization
- Three distinct failure modes require different interventions
- Existing benchmarks report aggregate accuracy only — no error attribution



Why This Problem Matters

Real-World Stakes

Models with 128K–1M token windows are deployed for contract analysis, multi-document synthesis, and codebase navigation — yet frontier models achieve only 50.1% accuracy on LongBench v2 vs. a 53.7% human baseline.

The Hidden Gap

Prior benchmarks report aggregate accuracy but cannot distinguish a model that misses the passage from one that misreads it — two failure modes demanding fundamentally different fixes.

Our Contribution

A diagnostic framework with a validated three-category error taxonomy (RF · RsF · INC), with high inter-annotator agreement under a deterministic decision protocol, enabling precise attribution of where and why LLMs fail on long-context tasks.

Building on Prior Benchmarks

Aggregate Accuracy Only

LongBench v2 & RULER establish that models fall short of human comprehension — but report only overall accuracy with no per-failure attribution.

Realistic Failure Modes

NIAH tests show near-perfect synthetic retrieval, but Lost in the Haystack and Context Rot reveal accuracy drops with real distractors and small target spans.

Validated Error Taxonomies

HELMET calls for instruction-adherence evaluation, yet provides no annotated taxonomy with inter-annotator reliability statistics.

What Our Framework Adds

Structured error annotation with high inter-annotator agreement under a deterministic protocol, disentangling **Retrieval Failure (RF)**, **Reasoning Failure (RsF)**, and **Instruction Non-Compliance (INC)**—the diagnostic layer prior benchmarks lack.

Extract-Then-Answer (E2A)

Extract Evidence



Answer Question

Key Idea

Break long-context reasoning into sequential, interpretable steps to better isolate where errors occur

Systematically evaluate whether introducing a structured decomposition improves overall accuracy.

8K Direct Answer

32K E2A

Single Step

Two-Stage

Short Context

Long Context

Pilot Study Configuration

Dataset & Model

LongBench v2 - 503 instances, 6 categories, contexts up to 2M words.

Pilot: stratified 50-instance sample (8–9 per category).

Llama-3.1-8B-Instruct - 128K nominal window, deterministic decoding (temperature = 0).

Conditions

8K DA: Direct Answer, ~3.8k avg input tokens

32K E2A: Extract-Then-Answer, ~30,7k avg input tokens

Error Taxonomy

RF - Retrieval Failure

No supporting evidence in context

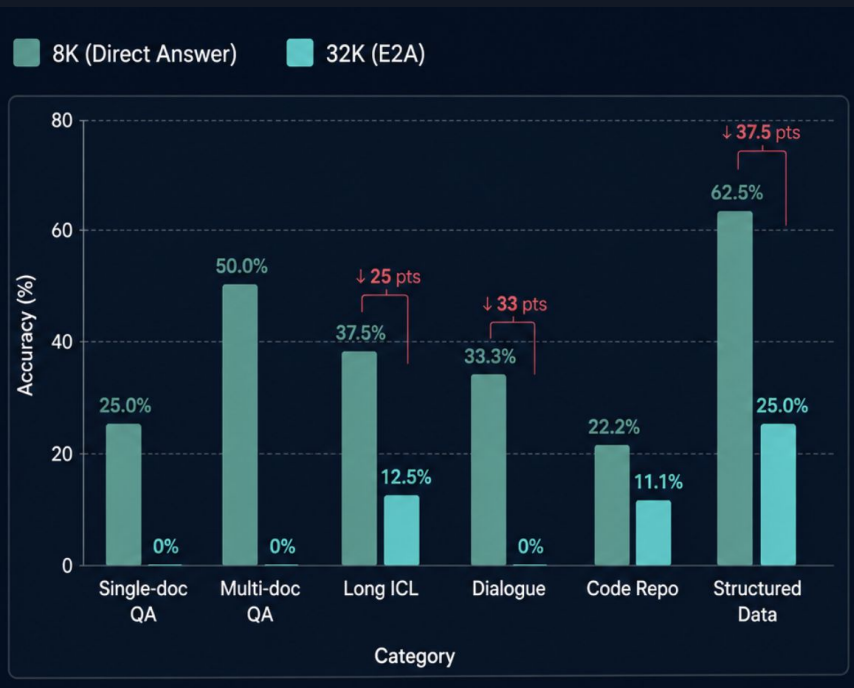
RsF - Reasoning Failure

Evidence present; wrong inference drawn

INC - Non-Compliance

Output not a valid A/B/C/D letter

Accuracy by Budget and Category



Key Observations

- Overall accuracy declines from 26.0% (8K DA) to 20.0% (32K E2A) despite more context
- 80% of E2A Stage 1 extractions return the sentinel — Stage 2 guesses from empty evidence
- Single- and multi-doc QA improve at 32K: longer context lets answer spans survive truncation
- Structured data collapses from 62.5% → 25.0%: E2A fragments table structure
- Latency increases 45% (3,200 → 4,635 ms) for a net accuracy loss

What the Error Analysis Reveals

65%
INC at 32K

Instruction Non-Compliance dominates at 32K - blank outputs from failed Stage 1 extraction

50%
RsF at 8K

Reasoning Failure dominates at 8K - model finds evidence but draws wrong conclusions

5%
RF at Both

Retrieval Failure is rare at both budgets, challenging the assumption that long-context failures are primarily retrieval problems

Inter-Annotator Agreement

Consistent labeling across 44 annotated instances—supporting a reliable and well-defined error taxonomy

Challenges & Open Questions

Pilot Scale

50 instances (20 annotated per budget); full 503-instance error breakdown pending

E2A Brittleness

Stage 1 sentinel rate of 80% suggests the extraction prompt itself may be underspecified — few-shot examples not yet tested

Single Model

Results limited to Llama-3.1-8B-Instruct; model-size effects not yet separated from prompting effects

Fixed Token Budgets

Only two points (8K, 32K) on the accuracy-vs-length curve; 64K and 128K needed to characterize the full scaling profile

What We Found

01

Bigger Context $\neq$ Better Accuracy

Moving from 8K DA to 32K E2A dropped accuracy from 26% -> 20%. More tokens hurt when the extraction mechanism fails 80% of the time.

02

INC Dominates, Not Retrieval Failure

Instruction Non-Compliance accounts for 45-65% of errors across both budgets. RF was only 5% - models aren't missing passages, they're formatting incorrectly.

03

Prompting Strategy $\times$ Task Type

Structured data drops 62.5% -> 25% under E2A. Single/multi-doc QA improves. Strategy choice must match task type - there is no universal winner.

Five Next Steps

1

Scale to 64K & 128K

Plot full accuracy vs length curves and compute degradation slopes across all six categories.

2

E2A Prompt Engineering

Add few-shot extraction examples and structured output constraints to cut the 80% sentinel failure rate.

3

Needle-Position Perturbation

Use BM25 span detection to move answer spans to start / middle / end - isolate positional retrieval bias.

4

Expand Annotation

Annotate all incorrect predictions (not just 20/budget) for per-category error breakdowns.

5

Larger & Frontier Models

Compare Llama-3.1-70B, GPT-4o, Claude 3.5 Sonnet separate model-size from prompting strategy effects.

Thank You

Questions?

26%

8K DA Accuracy

20%

32K E2A Accuracy

80%

Stage 1 Sentinel Rate